

Clinical Risk Assessment Report: Patient 48

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 36.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.63x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +7.1%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT)	Value: -0.35 [Bottom 12%] (+7.1%)
<i>Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.</i>	
Squamous Cell Carcinoma Antigen (SCC_Ag)	Value: 0.7 [Avg (28th %ile)] (+2.9%)
<i>Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.</i>	
Small Zone Emphasis (PET) (GLZLM_SZE.PET)	Value: 0.99 [Top 15%] (+2.7%)
<i>Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.</i>	
75th Percentile Tumor Density (CONVENTIONAL_HUQ3)	Value: 0.61 [Avg (64th %ile)] (+1.5%)
<i>Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.</i>	
Tumor Pixel Correlation (CT) (GLCM_Correlation.CT)	Value: -0.73 [Bottom 22%] (+1.4%)
<i>Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.</i>	
Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT)	Value: -0.22 [Bottom 15%] (+1.4%)
<i>Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.</i>	
Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET)	Value: 0.45 [Avg (64th %ile)] (+0.4%)
<i>Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.</i>	
Minimum Tumor Density (CONVENTIONAL_HUmin)	Value: 0.64 [Top 25%] (+0.3%)
<i>Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.</i>	

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

